Bitcoin Trader Performance vs Market Sentiment — Analysis Report

Objective: Analyze how Bitcoin market sentiment (Fear, Neutral, Greed) influences trader behavior and profitability using historical trading data and the Fear & Greed Index.

Methodology: Data from the Fear & Greed Index was merged with historical trade records. Features such as trade size (USD), leverage, long/short positions, and win/loss outcomes were engineered. Exploratory analysis was performed using Python (Pandas, Matplotlib, Seaborn). A Random Forest classifier was trained to predict profitable trades based on trading characteristics and market sentiment.

Metric	Fear	Neutral	Greed
Win Rate	40.8%	39.7%	42.0%
Mean PnL / Trade	$47.8	$33.3	$52.9
Avg Trades / Day	174	79	189

Key Insights:
- Trader profitability is highest during **Greed** market conditions.
- Neutral sentiment periods show the lowest trading activity and profitability.
- Low leverage traders achieve higher mean PnL compared to high leverage traders.
- Important predictive features for profitable trades include historical win rate, Fear & Greed score, coin trading frequency, and leverage.

Tools & Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Jupyter Notebook.

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